1.	**Prerequisites**	A knowledge of P1 and P2 and associated formulae and of vectors in two dimensions.
2.	**Examination**	• First assessment: June 2019. • The assessment is 1 hour and 30 minutes. • The assessment is out of 75 marks. • Students must answer all questions. • Calculators may be used in the examination. Please see *Appendix 6: Use of calculators*. • The booklet *Mathematical Formulae and Statistical Tables* will be provided for use in the assessments.
3.	**Notation and formulae**	Students will be expected to understand the symbols outlined in *Appendix 7: Notation*. Formulae that students are expected to know are given below and will **not** appear in the booklet *Mathematical Formulae and Statistical Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage. Momentum = mv Impulse = $mv - mu$ For constant acceleration: $v = u + at$ $s = ut + \frac{1}{2}at^2$ $s = vt - \frac{1}{2}at^2$ $v^2 = u^2 + 2as$ $s = \frac{1}{2}(u + v)t$

M1.3 Unit content

What students need to learn:		Guidance
1. Mathematical models in mechanics		
1.1	The basic ideas of mathematical modelling as applied in Mechanics.	Students should be familiar with the terms: particle, lamina, rigid body, rod (light, uniform, non-uniform), inextensible string, smooth and rough surface, light smooth pulley, bead, wire, peg. Students should be familiar with the assumptions made in using these models.